INCH-POUND

MIL-A-51487A(EA)
11 May 1990
SUPERSEDING
MIL-A-51487(EA)
1.4 April 1987

MILITARY SPECIFICATION

ARMING/FIRING UNIT

This specijica[ion is approved for use by the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Departmen t of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of arming/firing unit (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1040

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SPECIFICATIONS

MILITARY

MIL-W-63150 - Weapons and Support Materiel Standard Quality Assurance Provisions for

STANDARDS

FEDERAL

FED TEST METHOD STD. NO. 101 - Test Procedures for Packaging Materials

MILITARY

MIL-STD-454 - Standard General Requirements for Electronic Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other GOvernment documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

U.S. ARMY CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

13-12-111 - Arming/Firing Unit
13-12-118 - Nameplates

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P13-12-111 - Arming/Firing unit

(Copies are available from Commander, U.S. Army Information Systems Command, ATTN: ASQNC-ARI-CTF, Rock Island, IL 61229-7300.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing 13-12-111 or on subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristic set forth on the drawings.

3.1.2 Components. All components of the arming/firing unit shall conform to the specifications and drawings listed on Drawing 13-12-111 and subsidiary drawings.

3.2 Assembly. The arming/firing unit hall be assembled in accordance with Drawing 13-12-111.

3.3 Insulation resistance. The arming/firing unit shall have an insulation resistance of not less than 10 megohms when tested as specified in 4.4.5.3,

3.4 Continuity. The continuity of the arming/firing unit shall comply with the requirements of table I when tested as specified in 4.4.5.1.

3.5 Leakage. The arming/firing unit shall show no evidence of leakage when tested as specified in 4.4.5 .2.

3.6 First article. Prior to the start of regular production, a first article sample of arming/firing units shallbe produced (see. 6.2) in accordance with this specification for examination and tests (see 4.3).

3.7 Workmanship. The arming/firing unit shall be free from contamination (oil, rust, grease or foreign matter) and damage such as dents, cracks or deformation which could affect its intended use, and the arming/firing unit shall comply with the workmanship requirements of MIL-STD-454 and MIL-w-63150.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections Set forth in this specification where SUCh inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5 The inspection Set forth in this specification shall become a part of the contractor's

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overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

TABLE I. Continuity requirements - electrical resistance

Pin No.	Pin No.	ON/OFF arming switch position		
		Arm switch OFF	Arm switch ON Fire switch not depressed	Arm switch ON Fire switch depressed
A to	B	#	Lamp resistance	Lamp resistance
	C	#	#	0
	D	#	Lamp resistance	Lamp resistance
	E	#	#	0
	F	#	Lamp resistance	Lamp resistance
B to	C	#	#	Lamp resistance
	D	0	0	0
	E	#	#	Lamp resistance
	F	0	0	0
C to	D	#	#	Lamp resistance
	E	0	0	0
	F	#	#	Lamp resistance
D to	E	#	#	Lamp resistance
	F	0	0	0
E to	F	#	#	Lamp resistance

NOTES: 1. Letters .4 through F are connector pin identification

2. 0 indicates zero resistance
   # indicates infinite resistance
3. Lamp resistance - indicator light resistance varies from one light to another and therefore should be removed and checked prior to testing of unit. Normal resistances is 150 to 250 ohms.

4.1.2 Contractor assurance of compliance. The contractor's quality program or detailed inspection system shall provide assurance of compliance of all characteristics with the

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applicable drawing, special packnging instruction, and specification requirements using, as a minimum, the conformance criteria specified herein.

4.1.3 Alternaate inspection provisions. Alternate inspection procedures, methods, or equipment, such as statistical process control, tool control, and other types of sampling procedures may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the inspection provisions specified herein. Prior to applying such alternate procedures, methods, or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation and approval (see 6.3). When required, the contractor shrill demonstrate that the effectiveness of each proposed alternate is equal to or better than the quality assurance provisions specified herein. In cases of dispute as to whether the contractor's proposed alternate provides equal quality assurance, the provisions of this specification shall apply. All approved alternate inspection provisions shall be specifically incorporated into the contractor's quality program or detailed inspection system, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Qualify conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Sample. The first article sample shall. consist of five arming/firing units manufactured using the same methods, materials, equipment, and processes as will be used during regular production. However, when packaging other than for interplant shipment is required by the contract, the first article sample shall consist of five packaged arming/firing units and five empty barrier bags. The first article sample shall be submitted for inspection and approval in accordance with the terms of the contract.

4.3.2 Inspections to be performed. AS determined by the Government, the sample first article items may be subjected to any or all of the examinations and tests in this specification and be inspected for compliance with any or all of the requirements of the applicable drawings and special packaging instructions.

4.3.3 Rejection criteria. If any first article sample item fails to comply with any of the applicable requirements, the first article sample shall be rejected, The Government reserves the right to terminate inspection upon any failure to comply with any of the requirements. the contractor shall obtain written approval from the contracting activity prior to proceeding with regular production.

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4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the units offered for acceptance at one time which have been produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions, and at essentially the same time.

4.4.2 Sampling. Sampling shall be conducted in accordance with the classification of characteristics in 4.4.4 and, when specified, table II. Samples shall be selected at random. Empty barrier bags for the heat seal seam strength shall be selected at random from those being used for each lot of items.

TABLE II Sampling

Lot size	Inspection level					
	I	II	III	IV	V	VI
2 to 8	*	*	*	*	5	3
9 to 15	*	*	*	13	5	3
16 to 25	*	*	*	13	5	3
26 to 50	*	*	32	13	5	3
51 to 90	*	*	32	13	12	4
91 to 150	*	125	32	13	12	5
151 to 280	*	125	32	30	14	6
281 to 500	*	125	32	30	17	7
501 to 1200	*	125	74	35	20	9
1201 to 3200	1250	125	74	43	24	10
3201 to 10000	1250	125	87	50	30	10
10001 to 35000	1250	296	109	61	36	10
35001 to 150000	1250	296	124	74	40	10
150001 to 500000	1250	346	156	91	40	10
500001 and over	1250	431	187	102	40	10

Numbers under inspection levels indicate sample size; * indicates one hundred percent inspection. If sample size exceeds lot size, perform one hundred percent inspection.

Accept lot represented on zero and reject lot represented on one or more for all inspection levels.

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4.4.3 Inspection procedure. Sample items and, when applicable, the packaging thereof shall be examined and tested in accordance with the classification of characteristics in 4.4.4. Failure of sample items to conform to any characteristic in the classification of characteristics based on the sampling and acceptance criteria specified therein shall be cause for rejection of the lot represented.

4.4.4 Classification of characteristics. Quality conformance examinations and tests are specified in the following classification of characteristics paragraphs. When specified herein, accept on zero and reject on one attributes sampling inspection shall be performed on the designated characteristics using the stated levels in table 11 for the selection of sample sizes.

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.4(a)	Arming/Firing Unit		13-12-111
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical				
1	Continuity incorrect	100 percent	3.4	4.4.5.1
Major				
101	Leakage	Table II, level IV	3.5	4.4.5.2
102	Insulation resistance	Table II, level IV	3.3	4.4.5.3
103	Name plate locations or application incorrect	Table II, level IV	3.1	VI and CE
104	Threads damaged	Table II, level IV	3.1	VI
105	Screws loose or missing	Table II, level IV	3.1	VI
106	Protective finish missing or incorrect	Table II, level IV	3.1	VI
107	Workmanship	Table II, level IV	3.7	VI

NOTES:
CE - Commercial inspection equipment
VI - Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.4(b)	Name Plates		13-12-118
			NEXT HIGHER ASSY 13-12-111

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Dimensions incorrect	Table II, level IV	3.1	CE
102	Material incorrect	Table II, level IV	3.1	VI
103	Printing incorrect	Table II, level IV	3.1	VI

NOTES:
CE – Commercial inspection equipment
VI – Visual inspection

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CLASSIFICATION OF CHARACTERISTICS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.4(c)	Packaging		P13-12-111
			NEXT HIGHER ASSY

CATEGORY	CHARACTERISTIC	SAMPLING AND ACCEPTANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD
Critical	None defined			
Major				
101	Barrier bag leakage	Table II, level V	5.1	4.4.5.4
102	Heat seal seam strength (destructive test)	Table II, level V	5.1	4.4.5.5
103	Item not clean and dry prior to unit packing	Table II, level V	5.1	VI
104	Cap missing or incorrect	Table II, level V	5.1	VI
105	Bottom pad missing or incorrect	Table II, level V	5.1	VI
106	Inner container (box) missing or incorrect	Table II, level V	5.1	VI
107	Inner container (box) closure incorrect	Table II, level V	5.1	VI
108	Inner container (box) corners not blunted as specified	Table II, level V	5.1	VI
109	Barrier bag missing or incorrect	Table II, level V	5.1	VI
110	Stapled areas not covered by tape as specified	Table II, level V	5.1	VI
111	Outer container (box) missing or incorrect	Table II, level V	5.1	VI
112	Outer container (box) closure incorrect	Table II, level V	5.1	VI
113	Marking missing, incorrect, or illegible	Table II, level V	5.1	VI

NOTES:
VI – Visual inspection

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4.4.5 Tests. Tests shall be performed as follows:

4.4.5.1 Continuity. Using an ohmmeter determine resistance between each pin of connector. Resistance shall be as indicated in table I to be acceptable.

4.4.5.2 Leakage. The arming/firing unit shall be submerged in 3.0 ± 0.5 feet of water (914.40 millimeters) at ambient temperature for 15 minutes. Upon removal, the unit shall be thoroughly dried and the base plate removed. Any water found inside the unit constitutes leakage.

4.4.5.3 Insulation resistance. The insulation resistance between pairs of component circuits and between each circuit and ground shall be as specified in 3.3 when measured at 500 volts direct current (vdc) minimum, for a minimum of 30 seconds.

4.4.5.4 Barrier bag leakage. Determine the leakage of the barrier bag containing the arming/firing unit in accordance with the hot water technique of method 5009.2 of FED. TEST. METHOD STD. No. 101

4.4.5.5 Heat seal seam strength. Determine the heat seal seam strength of the empty barrier bag in accordance with method 2024 of FED. TEST. METHOD STD. No. 101 except that the static load shall be 45 (+5 or -0) ounces and a 5 percent reduction shall be permitted when the room temperature in the test area exceeds 90°F.

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in SPI P13-12-111.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful," but is not mandatory.)

6.1 Intended use. This item is a component of the M243 and M259 smoke grenade launchers.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required. the specific issue of individual documents referenced (see 2.1.1)
(c) First article information:
(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract when testing is performed by the Government.

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(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for the Government to notify the contractor whether or not to proceed with production.

6.3 Submission of alternate inspection provisions. Proposed alternate inspection provisions should be submitted by the contractor to the procuring contracting officer for evaluation and approval by the technical activity responsible for preparation of this specification.

6.4 Elimination of AQLs. This specification is certified to be in compliance with current Army Material Command (AMC) policy for the elimination of Acceptable Quality Levels/Lot Tolerance Percent Defective (AQLs/LTPDs) from military specifications.

6.5 Subject term (key word) listing.

Insulation resistance
Launcher, smoke grenade

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1040-A191

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-51487A(EA)	ARMING/FIRING UNIT

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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NOTICE OF INACTIVATION
FOR NEW DESIGN

INCH–POUND

MIL-A-51487A(EA)
NOTICE 1
27 December 1995

MILITARY SPECIFICATION

ARMING/FIRING UNIT

This notice should be filed in front of MIL-A-51487A(EA)
dated 11 May 1990.

MIL-A-51487A(EA) is inactive for new design and is no longer used except for replacement purposes.

Preparing activity:

Army - EA

AMSC N/A

FSC 1040

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